

Montage Wi-Fi M88WI6700 SDK Driver Compilation for Android 6.0 ver. 1.0.0

Release History

Version	Date	Remarks
1.0.0	2017/02/21	Initial Release

Contents

Release History.....	1
1. Overview.....	3
2. SDK Setup.....	3
2.1 SDK file and folder information.....	3
2.2 Copy The Files to The Related folders of the Platform.....	3
3. Platform Setup.....	4
3.1 Setup the WIFI_MODULE configuration.....	4
3.2 Setup the WiFi in the Project.....	4
3.3 Setup the Montage WiFi driver related configurations.....	4
3.4 Setup the wpa_supplicant setting.....	7
3.5 Setup the DHCP setting.....	7
3.6 System Resource Configurations.....	8
3.6.1 networkAttributes.....	8
3.6.2 radioAttributes.....	9
3.6.3 config_tether_wifi_regexs.....	9
3.6.4 config_tether_upstream_types.....	9
3.6.5 config_enableWifiDisplay.....	9
4. FAQ.....	10
4.1 STA Mode.....	10
4.1.1 wpa_supplicant can't work.....	10

1. Overview

This document provides the Montage WiFi Android SDK information to help the developers to create their project. And the m88wi6700 driver support the m88wi6700u (USB interface) and the m88wi6700s (SDIO) chips.

The SDK setup progress in this document is based on the Amlogic P212 platform. And there may be some difference between Amlogic's platform and the others. The developers should take care that and change the setup details to fit their platform.

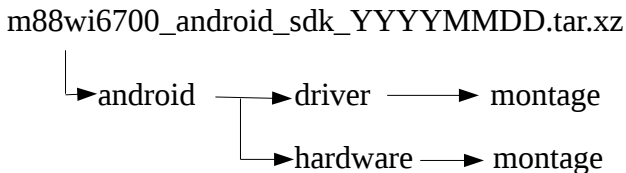
The Montage m88wi6700 driver is for the Linux Kernel 3.13 or 3.14. If the Kernel version of the Android is older than 3.13 or newer than 3.14, the Montage m88wi6700 driver could not work correctly.

2. SDK Setup

2.1 SDK file and folder information

The SDK file name will be m88wi6700_android_sdk_YYYYMMDD.tar.xz, ex: m88wi6700_android_sdk_20170210.tar.xz.

The files and folders architecture is:



The *driver/montage* folder contains the m88wi6700 driver source code. And the *hardware/montage* includes some configuration files and the private data.

2.2 Copy The Files to The Related folders of the Platform

The Amlogic P212 platform is used as the target platform. The developers should find out the folders location of the target platform. And the Amlogic example is as below:

The *driver/montage* of SDK is copied to *hardware/wifi/montage* of the Platform.

The *hardware/montage* of SDK is copied to *hardware/amlogic/wifi/montage* of the Platform.

3. Platform Setup

3.1 Setup the WIFI_MODULE configuration

path = ./device/<PLATFORM_VENDOR>/<PROJECT_NAME>/Kernel.mk

ex: ./device/amlogic/p212/Kernel.mk

Add the following comment into the Kernel.mk

```
WIFI_MODULE := <Montage Chip name>
```

for example:

If the chip is M88WI6700U (with usb interface), the setting is :

```
WIFI_MODULE := m88wi6700u
```

If the chip is M88WI6700S (with sdio interface), the setting is :

```
WIFI_MODULE := m88wi6700s
```

3.2 Setup the WiFi in the Project

path = ./device/<PLATFORM_VENDOR>/<PROJECT_NAME>/<PROJECT_NAME>.mk

ex: ./device/amlogic/p212/p212.mk

The following is the usb interface case. Modifying the *m88wi6700u* as *m88wi6700s* is for sdio chip.

```
#####  
#  
#                               WiFi  
#  
=====
```

```
WIFI_MODULE := m88wi6700u  
include device/amlogic/common/wifi.mk  
  
# Change this to match target country  
# 11 North America; 14 Japan; 13 rest of world  
PRODUCT_DEFAULT_WIFI_CHANNELS := 11
```

3.3 Setup the Montage WiFi driver related configurations

path = ./device/<PLATFORM_VENDOR>/common/wifi.mk

ex: ./device/amlogic/common/wifi.mk

Add the following comments into the wifi.mk for m88wi6700u chip

```
ifeq ($(WIFI_MODULE), m88wi6700u)
BOARD_WIFI_VENDOR      := montage
WIFI_DRIVER := m88wi6700u
WIFI_DRIVER_MODULE_PATH := /system/lib/m88wi6700u.ko
WIFI_DRIVER_MODULE_NAME := m88wi6700u
WIFI_DRIVER_MODULE_ARG := ""

BOARD_WLAN_DEVICE := m88wi6700u
WIFI_DRIVER_FW_PATH_PARAM := ""

WPA_SUPPLICANT_VERSION := VER_0_8_X
BOARD_WPA_SUPPLICANT_DRIVER := NL80211
BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_montage
BOARD_HOSTAPD_DRIVER      := NL80211
BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_montage

PRODUCT_PACKAGES += wpa_supplicant.conf \
                    wpa_supplicant_overlay.conf \
                    p2p_supplicant_overlay.conf

PRODUCT_COPY_FILES +=
frameworks/native/data/etc/android.hardware.wifi.xml:system/etc/permissions/android.hardware.
wifi.xml
PRODUCT_COPY_FILES +=
frameworks/native/data/etc/android.hardware.wifi.direct.xml:system/etc/permissions/android.hard
ware.wifi.direct.xml
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/wpa_supplicant.conf:system/etc/wifi/wpa_supplicant.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/wpa_supplicant_overlay.conf:system/etc/wifi/wpa_suppl
icant_overlay.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/p2p_supplicant_overlay.conf:system/etc/wifi/p2p_suppl
icant_overlay.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/tools/mont_ctrl:system/bin/mont_ctrl

PRODUCT_PROPERTY_OVERRIDES += wifi.interface=wlan0

endif
```

Add the following comments into the wifi.mk for m88wi6700s chip

```
ifeq ($(WIFI_MODULE), m88wi6700s)
BOARD_WIFI_VENDOR      := montage
WIFI_DRIVER             := m88wi6700s
WIFI_DRIVER_MODULE_PATH := /system/lib/m88wi6700s.ko
WIFI_DRIVER_MODULE_NAME := m88wi6700s
WIFI_DRIVER_MODULE_ARG  := ""

BOARD_WLAN_DEVICE := m88wi6700s
WIFI_DRIVER_FW_PATH_PARAM := ""

WPA_SUPPLICANT_VERSION := VER_0_8_X
BOARD_WPA_SUPPLICANT_DRIVER := NL80211
BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_montage
BOARD_HOSTAPD_DRIVER      := NL80211
BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_montage

PRODUCT_PACKAGES += wpa_supplicant.conf \
                    wpa_supplicant_overlay.conf \
                    p2p_supplicant_overlay.conf

PRODUCT_COPY_FILES +=
frameworks/native/data/etc/android.hardware.wifi.xml:system/etc/permissions/android.hardware.wifi.xml
PRODUCT_COPY_FILES +=
frameworks/native/data/etc/android.hardware.wifi.direct.xml:system/etc/permissions/android.hardware.wifi.direct.xml
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/wpa_supplicant.conf:system/etc/wifi/wpa_supplicant.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/wpa_supplicant_overlay.conf:system/etc/wifi/wpa_supplicant_overlay.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/config/p2p_supplicant_overlay.conf:system/etc/wifi/p2p_supplicant_overlay.conf
PRODUCT_COPY_FILES +=
hardware/amlogic/wifi/montage/tools/mont_ctrl:system/bin/mont_ctrl

PRODUCT_PROPERTY_OVERRIDES += wifi.interface=wlan0

endif
```

3.4 Setup the wpa_supplicant setting

path =

./device/<PLATFORM_VENDOR>/<PROJECT_NAME>/init.<PLATFORM_VENDOR>.board.rc

ex: ./device/amlogic/p212/init.amlogic.board.rc

For using the normal wifi function, add the following comment into the init.amlogic.board.rc :

```
service wpa_supplicant /system/bin/wpa_supplicant \
-iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
-I/system/etc/wifi/wpa_supplicant_overlay.conf \
-O/data/misc/wifi/sockets \
-e/data/misc/wifi/entropy.bin -g@android:wpa_wlan0
class main
socket wpa_wlan0 dgram 660 wifi wifi
disabled
oneshot
```

Add the following comment into the init.amlogic.board.rc to enable P2P function :

```
service p2p_supplicant /system/bin/wpa_supplicant \
-iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
-I/system/etc/wifi/p2p_supplicant_overlay.conf \
-p use_p2p_group_interface=1p2p_device=1 \
-O/data/misc/wifi/sockets \
-e/data/misc/wifi/entropy.bin -g@android:wpa_wlan0
class main
socket wpa_wlan0 dgram 660 wifi wifi
disabled
oneshot
```

3.5 Setup the DHCP setting

path =

./device/<PLATFORM_VENDOR>/common/products/<PRODUCT_NAME>/init.<PLATFORM_VENDOR>.rc

ex: ./device/amlogic/common/products/mbox/init.amlogic.rc

Add the following the comments to setup the wifi dhcp :

```
service dhcpcd_wlan0 /system/bin/dhcpcd -ABKL
    class main
    disabled
    oneshot

service iprenew_wlan0 /system/bin/dhcpcd -n
    class main
    disabled
    oneshot
```

If the P2P is enabled, additionally add the follows :

```
service dhcpcd_p2p /system/bin/dhcpcd -aABKL
    class main
    disabled
    oneshot

service iprenew_p2p /system/bin/dhcpcd -n
    class main
    disabled
    oneshot
```

3.6 System Resource Configurations

path =

./device/<PLATFORM_VENDOR>/<PROJECT_NAME>/overlay/frameworks/base/core/res/res/values/config.xml

ex: ./device/amlogic/p212/overlay/frameworks/base/core/res/res/values/config.xml

or

path = ./frameworks/base/core/res/res/values/config.xml

3.6.1 networkAttributes

This is used by the connectivity manager to decide which networks can coexist based on the hardware.

```
<string-array translatable="false" name="networkAttributes">
  <item>"wifi,1,1,1,-1,true"</item>
  <item>"ethernet,9,9,9,-1,true"</item>
  <item>"pppoe,8,8,1,-1,true"</item>
</string-array>
```

3.6.2 radioAttributes

This is used by the connectivity manager to decide which networks can coexist based on the hardware.

```
<string-array translatable="false" name="radioAttributes">
  <item>"1,1"</item>
  <item>"9,1"</item>
  <item>"18,1"</item>
```

3.6.3 config_tether_wifi_regexs

Setup the tetherable Wifi interfaces.

```
<string-array translatable="false" name="config_tether_wifi_regexs">
  <item>"wlan0"</item>
</string-array>
```

3.6.4 config_tether_upstream_types

Setup the allowable type (1 : TYPE_WIFI) for tethering.

```
<integer-array translatable="false" name="config_tether_upstream_types">
  <item>1</item>
  <item>8</item>
  <item>9</item>
</integer-array>
```

3.6.5 config_enableWifiDisplay

Support the WiFi Display

```
<bool name="config_enableWifiDisplay">true</bool>
```

4. FAQ

4.1 STA Mode

4.1.1 wpa_supplicant can't work

- The driver must exist : /sys/lib/m88wi6700u.ko or /sys/lib/m88wi6700s.ko.
- Check the wpa_supplicant.conf path and name in the **service wpa_supplicant** setting, and make sure that the file has existed. For example :

```
service wpa_supplicant /system/bin/wpa_supplicant \  
-iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \  
.....
```

The /data/misc/wifi/wpa_supplicant.conf must be existed.